

BA3-T4 - Provenance, identity/lifecycle and derivation/change-impact closure review

Revision: R1

Status: CLOSED / PASS / BA3 CLOSED

Repository baseline reviewed: a20a590d771b1053ecbd095d8a8c4255ae762bfe

Phase: BA3 - provenance, derivation, identity/lifecycle and change-revalidation mechanics

BA0: CLOSED

BA1: CLOSED

BA2: CLOSED

BA4: NOT STARTED

1. Closure question

BA3-T4 asks only:

Can the combined BA3-T1/T2/T3 contract be reduced, merged or simplified without losing source drill-down, derivation reproducibility, uncertainty localization, cross-baseline identity history, targeted revalidation or the governed-document authority boundary?

The review is adversarial closure, not vocabulary discovery. No new BA3 mechanism is accepted unless removal/collapse pressure exposes a concrete material counterexample.

2. Evidence reviewed

The closure review reuses the controlled evidence already established by the BA3 sequence:

1. BA0 responsibility/non-goals and governed-document authority boundary;
2. BA1 closed **BAReferent** + **BAProposition** identity ontology;
3. BA2 closed proposition/operator/role semantics;
4. BA3-T1 provenance/origin lower bound;
5. BA3-T2 M1/M2/M3 and order/WMS identity/lifecycle pressure;
6. BA3-T3 M4, provider-state normalization and feedback-authority pressure;
7. diagnostic resolution across governed correction.

No new method-specific ontology, STRIDE schema, ThreatForge runtime model or arbitrary implementation requirement is introduced.

3. Reduction review - provenance and baseline scope

3.1 Remove provenance from one BA1 identity family

Attempt: keep proposition provenance only and infer referent provenance from participating propositions, or vice versa.

Result: REJECTED.

A referent can survive while propositions about it change, and a proposition can cite a clause whose participating referents are supported across different source branches. Removing independent provenance from either family makes source drill-down and lifecycle review depend on indirect reconstruction.

Disposition: PASS / INDEPENDENT PROVENANCE RETAINED.

3.2 Remove immutable governed-baseline scope

Attempt: identify source evidence by document identity + locator only.

Result: REJECTED.

M1 directly falsifies this reduction: D-3.5 can remain the recognizable Decision area while its realization changes from Ethernet to Wi-Fi. A baseline-free source pointer can conflate materially different governed meanings.

Disposition: PASS / BASELINE SCOPE RETAINED.

3.3 Replace exact locator with copied source prose

Attempt: copy source text into BA and treat the copy as evidence.

Result: REJECTED.

The copy can drift and would create a second authority. Exact drill-down to the governed baseline remains the semantic lower bound; cached excerpts remain non-authoritative display aids.

Disposition: PASS / GOVERNED LOCATOR RETAINED.

4. Reduction review - source, derivation and context links

4.1 Merge `sourceLink` and `derivationBasisBinding`

Attempt: use one undifferentiated `evidence` relation.

Result: REJECTED.

Direct governed support and analytical derivation input answer different authority questions. A derived classification may use an accepted BA proposition as basis, but that proposition is not itself the project authority for the classification. Conversely, a grounded transfer needs direct source support without being analytically derived.

Disposition: PASS / SEMANTIC SEPARATION RETAINED.

4.2 Merge `revalidationContext` into `sourceLink`

Attempt: treat every context that can invalidate an element as direct source provenance.

Result: REJECTED by M4.

The sibling representation Decision must make the old capture-transfer meaning stale, but it does not directly ground a successor `transfer(... OpaqueEvidenceReference)` proposition. Promoting it to source authority would permit silent BA repair from context rather than governance.

Disposition: PASS / REVALIDATION CONTEXT RETAINED.

4.3 Merge `revalidationContext` into `derivation basis`

Attempt: treat effective context as an analytical derivation input.

Result: REJECTED.

M4's old transfer is grounded in FR-3.4; the sibling Decision affects its validity but does not make the proposition derived. The same problem appears wherever a grounded assertion's applicability depends on a separately governed context.

Disposition: PASS / CONTEXT IS NOT DERIVATION.

4.4 Promote revalidationContext to BA2 dependOn

Attempt: reuse the project-semantic dependency operator.

Result: REJECTED.

A revalidation trigger does not assert that the project-semantic target requires the context as a domain prerequisite. Doing so would invent project meaning merely to support analysis bookkeeping.

Disposition: PASS / BA2 REMAINS CLOSED.

5. Reduction review - origin, review and freshness dimensions

5.1 Collapse GROUNDED and DERIVED

Attempt: treat every accepted BA element as simply source-backed.

Result: REJECTED.

Provider normalization and method-neutral classification require explicit analytical transformation. Collapsing origin would allow tool/analyst inference to masquerade as directly governed project truth.

5.2 Collapse DIAGNOSTIC_UNRESOLVED into REJECTED

Attempt: treat unresolved meaning as a failed candidate.

Result: REJECTED.

A localized contradiction/ambiguity can itself be an accepted analytical diagnostic even though no project-semantic answer is accepted. M4 can require exactly this outcome when sibling Decision and FR remain inconsistent.

5.3 Remove reviewState

Attempt: infer acceptance from presence in the BA store.

Result: REJECTED.

Tool/analyst candidates and replacement proposals require a state before acceptance, and rejected candidates may remain auditable without entering accepted BA meaning.

5.4 Collapse freshness into reviewState

Attempt: use PENDING_REVIEW as the only marker for both new candidates and impacted prior meaning.

Result: REJECTED.

A new unreviewed candidate and a previously accepted meaning made suspect by a target-baseline change are different analytical conditions. Historical acceptance for B0 must coexist with staleness relative to B1; freshness therefore remains a distinct semantic dimension even if an implementation computes or co-locates it.

Disposition for Section 5: PASS / ORIGIN != REVIEW != FRESHNESS.

6. Reduction review - derivation reproducibility

6.1 Return to untyped derivationBasis

Attempt: record only an unordered set of basis references.

Result: REJECTED by provider normalization.

The raw provider vocabulary and the governed project mapping are not interchangeable inputs. The rule must know which basis item fills which input responsibility.

6.2 Allow mutable/unversioned rule references

Attempt: reference a logical rule key whose meaning can change in place.

Result: REJECTED.

The same basis could produce a different accepted result while historical provenance still points to the same rule name. Exact immutable rule revision is therefore necessary for replay.

6.3 Require a universal executable derivation language

Attempt: close BA3 only if every derivation can be expressed in one executable DSL.

Result: REJECTED AS UNNECESSARY.

Current evidence requires inspectable input roles, applicability, output semantics and normative rationale. Semantic replay is sufficient; byte-identical/tool-identical execution is not a BA0 responsibility.

6.4 Require exhaustive derivation-rule enumeration before closure

Attempt: predefine every future rule key.

Result: REJECTED AS UNNECESSARY.

The descriptor contract is closed; the inventory remains corpus-driven and extensible. A new rule key that satisfies the closed descriptor does not by itself reopen BA3.

Disposition: PASS / ROLE-BOUND BASIS + IMMUTABLE INSPECTABLE RULE IS THE CURRENT LOWER BOUND.

7. Reduction review - continuity and lifecycle

7.1 Use one identity-continuity rule for referents and propositions

Attempt: retain/replace both families under one generic textual or source-based equivalence test.

Result: REJECTED.

M1 preserves `LocalConnectivity` referent identity while replacing the Ethernet realization proposition. M3 can preserve capture identity while retiring remote-transfer assertion identity. Referent reuse and assertion identity therefore have materially different continuity tests.

7.2 Copy source-layer revise into BA lifecycle

Attempt: add or reuse a BA `REVISE` state whenever a governed document is revised.

Result: REJECTED.

One source revision can cause some BA identities to retain, others to replace and others to retire. Source revision is not BA semantic identity disposition.

7.3 Merge REPLACE and RETIRE

Attempt: use one generic CHANGE/END disposition.

Result: REJECTED.

M1 realization change has a semantic successor; M3 remote transfer can simply cease to exist. Successor identity versus no successor is analytically material and must remain explicit.

7.4 Remove continuityBasis

Attempt: store only the disposition and successor link.

Result: REJECTED.

Cross-baseline identity decisions would become opaque precisely when source wording, realization or document location changes while semantic identity is retained. Baseline-scoped evidence/rationale is required for reproducible continuity review.

7.5 Store SUPERSEDED and RETIRED as independent lifecycle state machines

Attempt: create another first-class status dimension.

Result: REJECTED AS REDUNDANT.

They are historical interpretations of accepted REPLACE and RETIRE. No separate semantic family/state machine is required.

Disposition: PASS / RETAIN | REPLACE | RETIRE WITH FAMILY-SPECIFIC IDENTITY RULES RETAINED.

8. Cross-corpus localized-change regression

8.1 Facial M1 - Ethernet -> Wi-Fi

Expected closed behavior:

LocalConnectivity	RETAIN
WiredEthernet	RETIRE if unused elsewhere
WiFi	INTRODUCE
realize(LocalConnectivity, WiredEthernet)	REPLACE -> WiFi successor

Only directly sourced/context-linked semantics are candidates for revalidation. A transport-realization change does not by itself invalidate unrelated transfer or responsibility assertions, and it does not force BA-wide staleness.

Result: PASS.

8.2 Facial M2 - external -> project-owned transport

The negative responsibility proposition is replaced by affirmative project responsibility; external service consumption retires if consumption ceases; new transport obligations may be introduced. Abstract connectivity and unrelated consumer meaning can retain.

Result: PASS.

8.3 Facial M3 - remote -> local recognition

The remote transfer proposition retires; it is not rewritten as a local-processing assertion. Retirement/replacement of the **RecognitionProcessor** participant and explicit context links localize affected transport semantics. Unrelated project meaning is not invalidated globally.

Result: PASS.

8.4 Facial M4 - raw capture -> opaque reference

The sibling representation Decision is explicit `revalidationContext` for the affected old transfer meaning. When it changes, the old proposition becomes stale. If FR-3.4 remains inconsistent, BA emits/localizes `DIAGNOSTIC_UNRESOLVED`; it does not synthesize a successor project commitment from the sibling Decision.

Result: PASS / decisive evidence that direct source parentage alone is insufficient.

8.5 Order/WMS responsibility control

Internal inventory authority semantics retire/replace when responsibility moves to an external WMS. Service/contract/normalization meaning is introduced as required. Unrelated payment/fulfillment/order identities do not become stale without explicit source/basis/context/participant impact.

Result: PASS.

8.6 Provider-state normalization

A derived governed result uses role-bound raw-vocabulary and governed-mapping inputs under an immutable generic normalization rule. If either input changes, only dependent derived meaning becomes stale. Missing or contradictory mapping yields diagnostic/rejection rather than tool-private truth.

Result: PASS.

Disposition for Section 8: PASS / NO OVER-INVALIDATION OR UNDER-INVALIDATION COUNTEREXAMPLE SURVIVES CURRENT EVIDENCE.

9. Diagnostic resolution regression

A prior accepted `DIAGNOSTIC_UNRESOLVED` identity is historical analytical truth for its baseline.

After governed correction:

issue removed	-> old diagnostic RETIRE
materially different issue remains	-> old diagnostic REPLACE
new grounded/derived meaning	-> own origin/provenance/review

Retyping the old diagnostic in place as `GROUNDING` would falsify its historical origin.

Disposition: PASS.

10. Feedback-authority regression

The candidate chain remains:

```
governed docs B0
-> BA B0
-> downstream analysis
-> corrective documentation candidate
-> governed review
-> governed docs B1
-> BA B1
```

Two adversarial reductions fail:

1. using the analysis finding/candidate as `sourceLink` for BA@B1 - **REJECTED**, because analysis would become project authority;
2. patching BA directly after an accepted analysis decision before governed documentation changes - **REJECTED**, because the authoritative commitment would exist only in BA/tool state.

The accepted correction grounds the next BA only through governed documentation B1.

No `AnalysisRecord`, Finding identity or change-event identity is required inside BA3 to preserve this authority boundary.

Disposition: PASS / GOVERNED DOCUMENTATION REMAINS SOLE PROJECT AUTHORITY.

11. BA1 reopen test

The combined BA3 mechanics require semantic identity for project referents and analytical assertions, but not for a third BAE family.

Provenance attachments, review records, continuity records, rule descriptors, source references, baseline keys and revalidation contexts have metadata responsibilities. None demonstrates both BA1 split criteria: independent project-semantic identity plus reusable subtype-specific invariants that cannot be represented honestly outside a new BAE family.

Disposition: PASS / BA1 REMAINS CLOSED.

12. BA2 reopen test

BA3 uses metadata relations such as source, basis, continuity and revalidation context, but these are not project-semantic assertions.

No current case forces a BA2 `affects`, `derivedFrom`, `revalidates`, `supersedes` or lifecycle operator. Existing project meaning remains representable with the closed 13-key registry.

Disposition: PASS / BA2 REMAINS CLOSED.

13. Analytical-materialization identity check

T4 also asks whether `governedBaselineKey` is insufficient because derivation-rule revisions can evolve independently of project documentation.

Current evidence does not force a new first-class BA materialization identity: every accepted derived element records the exact immutable rule revision used, and the current thesis requires one accepted BA materialization per governed baseline/review context rather than concurrent semantically divergent accepted variants.

This remains an explicit reopen criterion: if a concrete workflow requires multiple concurrent accepted BA materializations for the same governed baseline and their semantic-registry/rule-version context cannot be recovered from element provenance, reopen only this BA3 baseline/materialization responsibility.

Disposition: PASS FOR CURRENT SCOPE / NO NEW BASELINE BAE FAMILY OR MANDATORY KEY FORCED.

14. Representation-independence/minimality check

Several contracts can be physically co-located without semantic loss:

- source/basis/context links may share one edge table with distinct link kinds;
- review/freshness may share one record;
- reverse source-to-BA navigation may be computed;

- lifecycle historical labels may be derived from accepted continuity dispositions.

What cannot be removed is the ability to recover the distinct semantic questions.

This is the closure-level interpretation of the project rule:

`semantic responsibility exists`

`!=`

`must become first-class metaclass / class / table / graph node`

Disposition: PASS / SEMANTIC CONTRACT FROZEN, PHYSICAL REPRESENTATION OPEN.

15. Closure decision

No material counterexample survives the integrated reduction/regression review.

The current evidence does not force:

- reopening BA1 or BA2;
- a new first-class BAE family;
- a new BA2 operator;
- a universal derivation DSL;
- a general dependency graph;
- whole-BA invalidation;
- analysis-output authority;
- or implementation-specific storage semantics.

The T1/T2/T3 responsibilities are therefore accepted as the current-scope lower bound, with the physical-combination freedom and reopen criteria recorded in `BA3_PROVENANCE_DERIVATION_LIFECYCLE_CHANGE_CONTRACT_R1.md`.

BA3-T1 COMPLETED / PROVISIONAL PASS

BA3-T2 COMPLETED / PROVISIONAL PASS

BA3-T3 COMPLETED / PROVISIONAL PASS

BA3-T4 CLOSED / PASS

BA3 CLOSED FOR CURRENT THESIS SCOPE

Next phase: BA4 projections. BA4 is not executed by this closure package.